Lecture 4: Limits III

Garrett Credi

09/17/2026

Quiz Time!

Let's get into formation for quiz time.

Last Time

We spent much of last lecture developing our ability to compute limits via:

1. Recognizing when functions were *continuous* so that we could reduce the computation of limits to simpler evaluations
2. Developing limit properties which allowed us to build up the computation of complicated limits via reduction to simpler ones

We also developed the notion of the *one sided limits* - restricted kinds of limits where we need only approach from a singular side.

Today's Goals

We'll apply both our new limit computation skills as well as our knowledge of one-sided limits to work with *limits at or involving infinity*.
We will also uncover our last tool to help us compute limits: *the Squeeze Theorem*.

Limits at/involving Infinity

Prelude I

Let's first examine one of the problems from the previous worksheet:
We had two functions whose graphs looked like:

And we wanted to compute $\lim_{x \to 1} g(f(x)) = \lim_{x \to 1} g((x-1)^2) = \lim_{u \to 0^+} g(u) = 3$

Prelude II

Key to that problem was noticing that:

$$x \to 1 \quad \iff \quad (x-1) \to 0^{+} \tag{1}$$

Which allowed us to do a “substitution” that made the limit easier to compute.
This will be our strategy for computing limits at $\pm\infty$! Since $\pm\infty$ is not an actual number we can compute with in a consistent/coherent way, we’ll try and replace $x \to \pm\infty$ with a limit as *some other quantity* approaches *some actual number*.

Limits at Infinity I

Before that, let's actually look at the literal definition to see why we want to avoid it:

Limits at Infinity

For a function $f(x)$, we say that $\lim_{x\to\infty} f(x) = L$ if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x > N$ for some real number N.
Similarly, we say that $\lim_{x\to-\infty} f(x) = L$ if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x < M$ for some real number M.

Do we want to prove things in this class like we would need to if we actually used this definition? NO!

Limits at Infinity II

Instead, let's see if we can relate it to something we could instead *compute* using our tools from last lecture.

$$x > N \iff 0 > x > N \tag{2}$$
$$\iff 0 < \frac{1}{x} < \frac{1}{N} \tag{3}$$
$$\iff 0 < \frac{1}{x} - 0 < \frac{1}{N} \tag{4}$$

In other words:

Limits at Infinity in terms of Finite Limits

Taking the limit as $x \to +\infty$ is *exactly equivalent* to taking the limit as $\frac{1}{x} \to 0^+$

Similarly $x \to -\infty$ is *exactly equivalent* to $\frac{1}{x} \to 0^-$!

Limits at Infinity III: Example

For example:

$$\lim_{x\to\infty}\frac{3x^2+2x+1}{2x^2+6x+131}=\lim_{\frac{1}{x}\to 0^+}\frac{3x^2+2x+1}{2x^2+6x+131} \tag{5}$$

$$=\lim_{u\to 0^+}\frac{3(\frac{1}{u})^2+2(\frac{1}{u})+1}{2(\frac{1}{u})^2+6(\frac{1}{u})+131} \tag{6}$$

$$=\lim_{u\to 0^+}\frac{u^2}{u^2}\frac{3\frac{1}{u^2}+2\frac{1}{u}+1}{2\frac{1}{u^2}+6\frac{1}{u}+131} \tag{7}$$

$$=\lim_{u\to 0^+}\frac{3+2u+u^2}{2+6u+131u^2} \tag{8}$$

$$=\lim_{u\to 0}\frac{3+2u+u^2}{2+6u+131u^2}=\frac{3+2\cdot 0+0^2}{2+6\cdot 0+131\cdot 0^2}=\frac{3}{2} \tag{9}$$

Limits at Infinity IV: Example 1+2

So we took a scary looking limit $\lim_{x\to\infty} \frac{3x^2+2x+1}{2x^2+6x+131}$ and reduced it to simply evaluating some rational function $f(u) = \frac{3+2u+u^2}{2+6u+131u^2}$ at $u = 0$!
What about for just a polynomial?

$$\lim_{x\to-\infty}(x^3+2x+1) = \lim_{u\to0^-}((\frac{1}{u})^3+2\frac{1}{u}+1) \tag{10}$$

$$= \lim_{u\to0^-}\frac{1}{u^3}(1+2u^2+u^3) \tag{11}$$

$$\text{“}=\text{”}(\lim_{u\to0^-}\frac{1}{u^3})\cdot(\lim_{u\to0^-}(1+2u^2+u^3)) \tag{12}$$

$$= \lim_{u\to0^-}\frac{1}{u^3} \tag{13}$$

Where we unfortunately encounter a problem since $\frac{1}{u^3}$ has an *asymptote* at $u = 0$ and so cannot be replaced by a function continuous at 0. What do we do?

Limits at Infinity V: Example 2 and Divergence to Infinity

Let's look at the graph!

Limits Diverging to Infinity

For a function $f(x)$ and a point a, we say that $\lim_{x \to a} f(x) = +\infty$ (read as the limit of $f(x)$ as x approaches a diverges to positive infinity) if, for any lower bound L we can force $f(x) > L$ just by forcing $x \approx a$ (up to some error $\delta > 0$).
Similarly, we say that $\lim_{x \to a} f(x) = -\infty$ (... diverges to minus infinity) if, for any upper bound U we can force $f(x) < U$ just by forcing $x \approx a$ (up to some error $\delta > 0$).

So, from the graph of $\frac{1}{u^3}$ as $u \to 0^-$ we see that $\lim_{u \to 0^-} \frac{1}{u^3}$ *diverges* to minus infinity.

Squeeze Theorem

Squeeze Theorem I

Let's consider one final limit:

$$\lim_{x \to \infty} \frac{\sin(x)}{x} = \lim_{u \to 0^+} \frac{\sin(\frac{1}{u})}{\frac{1}{u}} \tag{14}$$

$$= \lim_{u \to 0^+} u \sin(\frac{1}{u}) \tag{15}$$

And now we *REALLY* get stuck: as $u \to 0$, $\sin(\frac{1}{u})$ oscillates infinitely rapidly and escapes all ability to be pinned down to a value we could use to simplify the limit. Noooooo!
... ... but are we really stuck?

Squeeze Theorem II

How much does $\sin(\frac{1}{u})$ really affect the value of the limit? Since we know that $\sin$ as a function never gets large at all, intuitively it shouldn't be able to outpace u shrinking towards 0. Formally, since $-1 \leq \sin(x) \leq 1$ (the range of $\sin(x)$ is $[-1, 1]$) we have both that:

$$\lim_{u \to 0^+} u \sin(\frac{1}{u}) \leq \lim_{u \to 0^+} u \cdot 1 \tag{16}$$

$$= \lim_{u \to 0^+} u = \lim_{u \to 0} u \tag{17}$$

$$= 0 \tag{18}$$

and that:

$$\lim_{u \to 0^+} u \sin(\frac{1}{u}) \geq \lim_{u \to 0^+} u \cdot -1 \tag{19}$$

$$= -\lim_{u \to 0^+} u = -\lim_{u \to 0} u \tag{20}$$

$$= -0 = 0 \tag{21}$$

Squeeze Theorem III

So if

$$0 \leq \lim_{u \to 0^+} u \sin(\frac{1}{u}) \leq 0 \tag{22}$$

we must have that $\lim_{u \to 0^+} u \sin(\frac{1}{u}) = 0$!
This style of argument comes from the following result:

The Squeeze Theorem

Suppose we have a function $f(x)$ and a point a so that:

1. we can find other functions $g(x), h(x)$ so that (around $x = a$) $g(x) \leq f(x) \leq h(x)$
2. and that $\lim_{x \to a} g(x) = \lim_{x \to a} h(x)$

Then we must have that $\lim_{x \to a} f(x)$ exists and is equal to the common value of $\lim_{x \to a} g(x) = \lim_{x \to a} h(x)$.

Tools for Algebraically Computing Limits

Let's recap our tools:

1. Evaluation for continuous functions: if $f(x)$ is continuous at $x = a$ then

$$f(a) \quad = \quad \lim_{x \to a^+} f(x) = \lim_{x \to a} f(x) = \lim_{x \to a^-} f(x) \tag{23}$$

2. Reduction to continuous functions: maybe the point at which you are taking the limit towards isn't in the domain but you can replace it with one and use tool #1
3. Limit Properties: You can rewrite your limit as a sum, product, quotient, or composite of other limits and reduce it to computing those
4. Inversion for Limits at infinity: $x \to \infty \iff \frac{1}{x} \to 0^+$, $x \to -\infty \iff \frac{1}{x} \to 0^-$
5. Squeeze Theorem: maybe you have a limit which involves a term that "shouldn't matter" intuitively and you can use the range of that term to put bounds that squeeze the value of the limit.

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!